

Hypothesis Testing & P-Values

Today's Big Question: When you see a difference in your data — a new button design that converts better, a new drug that seems to work — how do you know if it's *real*, or if you just got lucky with this particular sample?

Why This Matters

Hypothesis testing is the formal toolkit data analysts use to separate signal from noise. Without it, every observed difference looks meaningful, and you'd ship every "improvement" your data shows you — even the ones that are pure chance. This is the foundation of A/B testing, experiment design, and any claim that starts with "the data shows..."

By the End of Today, You Will Be Able To:

- Explain the null hypothesis and alternative hypothesis in plain English
- Understand what a p-value actually means (and the #1 way people misread it)
- Read a significance level (alpha) and decide whether a result is statistically significant
- Understand Type I and Type II errors and the tradeoff between them
- Run and interpret a basic t-test in Python
- Connect hypothesis testing directly to A/B testing in the real world

Real-World Use Case: An e-commerce team changes their "Buy Now" button from blue to green. Over a week, the green button gets a 2% higher conversion rate. Is green really better, or did it just happen to have a good week? Hypothesis testing gives a precise, repeatable answer instead of a gut feeling.

How to Introduce This Topic

Opening Line for the Class

Try this: "I'm going to flip a coin 10 times and get 7 heads. Is this coin rigged?" Most people's gut instinct is uncertain — 7 out of 10 doesn't feel obviously unfair, but it doesn't feel perfectly fair either. Hypothesis testing is the formal way to answer exactly this kind of question with a number, instead of a gut feeling.

Why This Topic Comes Right After Correlation

Yesterday we learned correlation can show two variables move together, but never proves causation. Today's topic is the tool that lets us test a specific claim rigorously — including, eventually, causal claims from controlled experiments. Hypothesis testing is the bridge between "I see a pattern" and "I can defend this pattern statistically."

The Core Idea in One Sentence

We assume nothing interesting is happening (the **null hypothesis**), then ask: "if that boring assumption were true, how surprising is the data I actually observed?" If the data would be very surprising under "nothing is happening," we reject that assumption.

A Courtroom Analogy to Use Out Loud

Think of a criminal trial. The default assumption is **innocent until proven guilty** — this is the null hypothesis. The prosecution must present evidence strong enough to overcome that default. If the evidence is overwhelming, the jury rejects "innocent" and convicts. If the evidence is weak or ambiguous, the jury does NOT declare innocence proven — they simply fail to reject it. This subtle point trips up almost every beginner: **failing to reject the null is not the same as proving it true.**

PART 1

Null Hypothesis vs. Alternative Hypothesis

Null Hypothesis (H0): The "nothing special is happening" claim. There is no effect, no difference, no relationship. This is the default, boring assumption we test against.

Alternative Hypothesis (H1 or Ha): The claim you're actually interested in. There IS an effect, a difference, or a relationship.

Real Examples

Scenario	Null Hypothesis (H0)	Alternative Hypothesis (H1)
New website button color	Conversion rate is the same for blue and green buttons	Conversion rate is different between blue and green buttons
New drug trial	The drug has no effect on blood pressure	The drug changes blood pressure
Average salary by department	Average salary is the same across departments	Average salary differs across departments
Coin fairness	The coin is fair (50% heads)	The coin is not fair

Talking point: notice the null hypothesis always states *equality* or *no effect* — it's specific and testable. The alternative is often broader ("different," not necessarily "better" or "worse") unless you have a directional theory going in. Ask the class to write H0 and H1 for their own example before moving on — this is the single most useful exercise for today's session.

PART 2

P-Values — What Are We Actually Measuring?

Definition: The p-value is the probability of seeing data at least as extreme as what you observed, *assuming the null hypothesis is true*. It is NOT the probability that the null hypothesis is true, and it is NOT the probability that your result is due to chance. Both of those are the single most common misreadings of a p-value — flag this explicitly.

Plain-English Translation

A small p-value says: "if nothing interesting were really going on, it would be very unusual to see data like this." That unusualness is what makes us suspicious that the null hypothesis is wrong.

The Decision Rule

P-Value	Common Interpretation	Decision (at $\alpha = 0.05$)
$p < 0.01$	Very strong evidence against H_0	Reject H_0 — highly significant
$p < 0.05$	Strong evidence against H_0	Reject H_0 — statistically significant
$0.05 \leq p < 0.10$	Weak / borderline evidence	Fail to reject H_0 (often reported as 'trend')
$p \geq 0.10$	Little to no evidence against H_0	Fail to reject H_0

Golden Rule: A p-value answers "how surprising is this data if nothing is going on?" It does not answer "how big or important is the effect?" and it does not answer "is the null hypothesis true?" Statistical significance and practical significance are different things — a tiny, meaningless effect can still produce a small p-value if the sample size is large enough.

Significance Level (Alpha) — Where Do We Draw the Line?

Alpha (α) is the threshold you choose *before* running the test — the cutoff p-value below which you'll declare a result "statistically significant." The most common choice in industry and academia is $\alpha = 0.05$ (a 5% threshold), but it's a convention, not a law of nature.

Why 0.05? And When to Use Something Else

- $\alpha = 0.05$ became standard largely by historical convention (Ronald Fisher, 1920s) — it is not a magic number, just a widely agreed-upon default
- $\alpha = 0.01$: use when false positives are costly — medical treatments, safety-critical systems
- $\alpha = 0.10$: sometimes used in early-stage exploratory analysis where missing a real effect is worse than chasing a false one
- In A/B testing at a company, the chosen alpha is a business decision: how often are you willing to roll out a "winning" variant that wasn't actually better?

The Critical Distinction: Alpha vs. P-Value

Students often confuse these two numbers because they're both small decimals. Alpha is the line in the sand you draw *in advance*. The p-value is the result you get *after* running the test. You compare the p-value to alpha to make a decision — you never set alpha after seeing the data, or you've biased your own experiment.

Talking point: ask the class "if your company let you pick alpha after seeing the p-value, what would go wrong?" Lead them to see that you could always nudge alpha just above whatever p-value you got — making every result "significant." This is why alpha must be locked in before the experiment.

PART 3

Type I and Type II Errors

Every hypothesis test can go wrong in one of two ways. Understanding both is essential for making good business and scientific decisions.

	H0 Is Actually True	H0 Is Actually False
You Reject H0	Type I Error (False Positive) — you claim an effect that isn't real. Probability = α	Correct decision — you correctly detected a real effect
You Fail to Reject H0	Correct decision — you correctly found no effect	Type II Error (False Negative) — you missed a real effect. Probability = β

Real-World Framing

- **Type I Error (False Positive):** A spam filter marks a real, important email as spam. You shipped a website redesign that you believed improved conversions, but it actually made no difference
- **Type II Error (False Negative):** A spam filter lets a real spam email through. You had a genuinely better website design but your test wasn't sensitive enough to detect it, so you abandoned it

Talking point: these two error types trade off against each other. Lowering alpha (e.g., from 0.05 to 0.01) reduces Type I errors but increases Type II errors, because you're now demanding stronger evidence before you'll ever reject H0. The only way to reduce both simultaneously is to collect more data — this is exactly why A/B tests need a minimum sample size before you trust the result.

Statistical Power — Can Your Test Even Detect the Effect?

Power = $1 - \beta$ = the probability your test correctly detects a real effect when one truly exists. A common target in industry experiments is **80% power**.

What Drives Power Up or Down

Factor	Effect on Power
Larger sample size	Increases power — more data means more ability to detect a true effect
Larger true effect size	Increases power — bigger differences are easier to detect
Lower variability in the data	Increases power — less noise makes the signal clearer
Stricter alpha (e.g., 0.01 instead of 0.05)	Decreases power — you're demanding more evidence, so weaker effects slip through

Talking point: this is the practical reason A/B tests sometimes need to run for weeks rather than days — the team needs enough sample size to hit adequate power. Running an underpowered test and concluding "no significant difference found" is one of the most common silent mistakes in industry — it often just means the test never had a real chance of detecting the effect in the first place.

Key distinction to repeat: "not statistically significant" does NOT mean "no effect exists." It may simply mean the test was underpowered to detect the effect that's actually there.

PART 4

Running a T-Test in Python

A **t-test** compares the means of two groups to see if they're significantly different from each other. It's the workhorse test behind most basic A/B test analysis.

Example: Did the New Checkout Page Increase Average Order Value?

```
import numpy as np
from scipy import stats

# Average order value (Rs) for each group
control_group = [450, 480, 460, 470, 455, 465, 475, 440, 462, 458]
treatment_group = [490, 510, 495, 505, 488, 500, 515, 480, 498, 502]

# Independent two-sample t-test
t_stat, p_value = stats.ttest_ind(control_group, treatment_group)

print(f"T-statistic: {t_stat:.2f}")
print(f"P-value: {p_value:.4f}")

alpha = 0.05
if p_value < alpha:
    print("Reject H0: there IS a statistically significant difference")
else:
    print("Fail to reject H0: no statistically significant difference detected")
```

How to Read the Output

- **T-statistic:** how many standard errors apart the two group means are — bigger absolute value means a bigger relative difference
- **P-value:** compare directly against your pre-chosen alpha (commonly 0.05)
- If $p < 0.05$ here, you'd conclude the treatment group's average order value is genuinely higher, not just higher by chance in this sample

PART 5

Connecting It All to A/B Testing

A/B testing is hypothesis testing applied directly to product and marketing decisions. Every A/B test you'll run in your job follows the exact same five steps:

Step	What You Do	Today's Concept Used
1	State what you're testing (e.g., button color, email subject line)	Defining the experiment
2	Write H0 (no difference) and H1 (there is a difference)	Null & Alternative Hypothesis
3	Decide your significance level and required sample size in advance	Alpha & Statistical Power
4	Run the experiment, collect data from both groups	Data collection
5	Run the test (e.g., t-test), compare the p-value to alpha, and decide	P-value & Decision Rule

A Common Real-World Trap: Peeking

Many teams check their A/B test results every day and stop the test as soon as they see " $p < 0.05$." This inflates the real Type I error rate far above 5%, because you're effectively running many small tests and taking the first lucky result. The fix: decide your sample size and test duration in advance, and don't stop early just because the p-value dipped below 0.05 on day 2.

AVOID THESE

Common Mistakes Students Make

Mistake 1: 'A p-value of 0.03 means there's a 3% chance the null hypothesis is true.'

Reality: No. The p-value is the probability of the observed data (or more extreme) given that H_0 is true — it says nothing directly about the probability that H_0 itself is true.

Mistake 2: ' $p = 0.06$ means there's definitely no effect.'

Reality: No. It means you failed to reject H_0 at the 0.05 level — it does not prove H_0 is true. The effect could be real but your test may be underpowered, or the true effect could genuinely be small.

Mistake 3: 'Statistically significant always means practically important.'

Reality: No. With a huge sample size, even a tiny, meaningless difference can produce a very small p-value. Always check the effect size, not just the p-value.

Mistake 4: 'I'll just keep checking my A/B test daily and stop once it's significant.'

Reality: This is 'p-hacking by peeking' — it dramatically increases your real false-positive rate. Decide sample size and duration in advance.

Mistake 5: 'My test wasn't significant, so the null hypothesis is proven true.'

Reality: Failing to reject H_0 is not the same as proving H_0 . You simply didn't find enough evidence against it — possibly because of low statistical power.

PRACTICE

Problems — Hypothesis Testing & P-Values

Problem 1: You run an A/B test on a new homepage and get $p = 0.02$ with alpha set at 0.05. What do you conclude? What's your decision?

Hint: $p = 0.02 < 0.05$, so you reject H_0 — the result is statistically significant. The new homepage's effect is unlikely to be due to chance alone.

Problem 2: Your colleague says "our test wasn't significant, so the new feature definitely doesn't help." What's wrong with this statement?

Hint: failing to reject H_0 doesn't prove H_0 is true. The test might be underpowered (too small a sample) to detect a real, smaller effect.

Problem 3: A team checks their A/B test results every morning and stops the test the moment p drops below 0.05. Why is this risky?

Hint: this is 'peeking' — it inflates the real false-positive rate well beyond the nominal 5%, because you're effectively running multiple tests and taking the luckiest one.

Problem 4: Explain, in your own words, the difference between a Type I error and a Type II error using a medical test example.

Hint: Type I = healthy patient incorrectly told they're sick (false positive). Type II = sick patient incorrectly told they're healthy (false negative).

Problem 5: Write the null and alternative hypotheses for: "Does changing our email subject line increase open rates?"

Hint: H_0 : open rate is the same for both subject lines. H_1 : open rate differs between the two subject lines.

One-Tailed vs. Two-Tailed Tests

Type	When to Use	Example
Two-Tailed Test	You only care whether there IS a difference, in either direction	Is the new design's conversion rate different from the old one (higher OR lower)?
One-Tailed Test	You specifically care about a difference in ONE direction only	Is the new design's conversion rate higher than the old one? (you don't care if it's just 'different')

Talking point: one-tailed tests are more "powerful" for detecting an effect in the direction you specified, but they come with a catch — you must commit to that direction *before* seeing the data. Switching to a one-tailed test after peeking at results in the direction you wanted is a form of p-hacking. In most business A/B testing, two-tailed tests are the safer default.

```
from scipy import stats

# Two-tailed (default)
t_stat, p_value = stats.ttest_ind(control_group, treatment_group)

# One-tailed: testing if treatment_group mean is GREATER than control_group mean
t_stat, p_value_two_tailed = stats.ttest_ind(treatment_group, control_group)
p_value_one_tailed = p_value_two_tailed / 2 # only valid if t_stat direction matches your hypothesis
```

Cheat Sheet — Thursday

Concept	Definition	Key Point
Null Hypothesis (H_0)	The default 'no effect / no difference' claim	What we assume true until proven otherwise
Alternative Hypothesis (H_1)	The claim of an effect or difference	What you're trying to find evidence for
P-Value	Probability of data this extreme, IF H_0 were true	Smaller = more evidence against H_0 . NOT the probability H_0 is true
Alpha (α)	Significance threshold, chosen in advance	Common default: 0.05
Type I Error	False positive — rejecting a true H_0	Probability = α
Type II Error	False negative — failing to reject a false H_0	Probability = β
Power ($1 - \beta$)	Probability of detecting a real effect	Target: usually 80% or higher

Key Takeaways & Next Steps

Key Takeaways from Thursday

- Hypothesis testing is how we decide if an observed difference is real or just random noise
- The null hypothesis (H_0) assumes no effect; we look for strong enough evidence to reject it
- A p-value measures how surprising your data is *if H_0 were true* — it is not the probability H_0 is true
- Alpha is your significance threshold, chosen **before** the experiment — commonly 0.05
- Type I error = false positive; Type II error = false negative; they trade off against each other
- Statistical power depends on sample size, effect size, and variability — underpowered tests miss real effects
- A/B testing in industry is hypothesis testing in action — don't peek early, and decide your sample size in advance

Next Steps: Tomorrow (Friday), we'll bring everything together: Regression basics, an introduction to chi-square tests for categorical data, and a practical look at the SQL and dashboard tooling a working data analyst uses every day. This closes the loop from probability and correlation through testing and into the tools you'll actually use on the job.